

Mann Ahalpara

AI Engineer & Software Developer

AI & Data Science student ranked 1st in the ICT department, with expertise in Python, LLMs, and machine learning. Focused on applied AI research and end-to-end AI development, with a strong foundation in building and deploying intelligent systems using FastAPI, Gemini, and LangChain. Currently conducting research at NTU Singapore under Prof. Erik Cambria's Sentic Team, with a published paper at ICIDS 2025.

Contact Info:

Phone No: +91 9904445765
Email Id: mannahalpara@gmail.com
LinkedIn: <https://www.linkedin.com/in/mannahalpara/>
GitHub: <https://github.com/MannAhalpara>

Experience:

Research Intern (Remote): NTU Singapore May 2026 – Present

- Conducting research on privacy-preserving long-term Emotional Support Conversation (ESC) systems under the mentorship of Tiesunlong Shen and Prof. Erik Cambria's Sentic Team.

AI & ML Intern: Edunet Foundation (AICTE x IBM SkillsBuild) Jan 2026– Feb 2026

- Developed and deployed Study Buddy, an AI-powered learning assistant using Gemini API and Streamlit that provides topic explanations, note summarization, quiz generation, flashcards, and PDF-based learning support through document-processing and prompt-engineering workflows.

Education:

B.Tech in Information & Communication Technology | Adani

University, Ahmedabad | 2023 – Present

CGPA: 9.10/10 | Ranked 1st in the ICT Department

Class XII (HSC) | Gujarat Secondary & Higher Secondary Education Board | 2023
Percentile: 97.03%

Projects:

AnswerIQ | FastAPI, Next.js, Gemini Flash, Sentence Transformers, Supabase [GitHub](#)

- Built an AI-powered answer-sheet evaluation system using Gemini Flash OCR, semantic analysis, and LLM-assisted scoring.
- Developed a hybrid evaluation framework combining concept coverage (50%), semantic similarity (30%), and keyword matching (20%) to generate explainable answer scores.
- Engineered a question-normalization and answer-mapping pipeline that correctly aligns answers with questions despite varying numbering formats and submission order.

Mini-RAG System | FastAPI, Qdrant, Gemini, Vercel, Render

[GitHub](#) | [Live](#)

- Built and deployed a Retrieval-Augmented Generation (RAG) application that enables users to query uploaded PDF and text documents through citation-grounded AI responses.
- Developed an end-to-end retrieval pipeline using Gemini embeddings, Qdrant vector search, semantic retrieval, and reranking to improve answer relevance and retrieval quality.

AirVision – AQI Prediction Platform | Python, ML, Streamlit [GitHub](#) | [Live](#)

- Built a machine learning pipeline for AQI prediction using pollutant measurements from multiple Indian cities and benchmarked 11 regression algorithms.
- Achieved best performance with CatBoost ($R^2 = 0.91$, MAE = 12, RMSE = 19) and deployed the model as an interactive Streamlit dashboard for prediction and analysis.

AXON – AI Desktop Assistant | Python, Gemini, LLaMA [GitHub](#)

- Developed a voice- and text-controlled desktop assistant capable of launching applications, executing commands, performing web searches, and answering user queries.
- Implemented a dual-LLM architecture using Gemini for command interpretation and screen analysis, and LLaMA for conversational question answering.

Blinkit Retail Analytics Dashboard [PowerBI](#) | [Kaggle](#)

- Built an interactive Power BI dashboard covering customer, order, marketing, and inventory analytics using multi-table retail datasets.
- Published the dataset and dashboard on Kaggle, receiving 350+ views and 100+ downloads within two months.

Publication:

Analyzing Indian Stock Markets through Correlations: Comparative Insights | ICIDS 2025 [Paper](#) | [CorrelationX](#)

- Co-authored and published a conference paper analyzing inter-sector stock market relationships using Pearson, Spearman, Kendall's Tau, and time-lagged correlation techniques.
- Extended the research into CorrelationX, a stock market analytics platform enabling multi-method correlation analysis and AI-generated investment insights.

Skills:

- Languages:** Python, SQL, Java
- ML & Data Science:** Regression, Classification, Feature Engineering, Model Evaluation,EDA, Predictive Analytics, PowerBI
- AI / LLMs:** LLMs, RAG, Prompt Engineering, AI Agents, Semantic Search,Vector Databases
- Frameworks & Tools:** FastAPI, Flask, Streamlit, Pandas, NumPy, Scikit-learn,Sentence Transformers, Git, Docker
- Databases & Deployment:** MySQL, PostgreSQL, Supabase, Qdrant, Vercel, Render